

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

Constraint-Based Network Connectivity Solution for a Remote Rural Branch Office

1. Problem Identification

The company needs to establish a reliable network connection for a remote rural branch where high-speed wired internet is unavailable. The network must support cloud applications, video conferencing, and secure communication with headquarters while keeping costs low.

2. Constraints

- **Location constraint:** Wired broadband is unavailable.
- **Bandwidth constraint:** Cloud applications and video conferencing require sufficient bandwidth.
- **Reliability constraint:** Business operations should continue even if the primary connection fails.
- **Cost constraint:** The solution must fit a limited budget.
- **Security constraint:** Communication with headquarters must be protected.
- **Scalability constraint:** The network should support additional employees and devices in the future.

3. Proposed Solution

Internet Connection

Use **4G/5G mobile broadband as the primary connection**, depending on network availability in the rural area.

For reliability, use a **secondary connection such as satellite internet or another mobile network provider** as backup. A dual-WAN router can automatically switch to the backup connection when the primary link fails.

Networking Devices

- **Dual-WAN Router:** Manages primary and backup internet connections and provides automatic failover.
- **Network Switch:** Connects computers, printers, IP phones, and other wired devices.
- **Wi-Fi Access Point:** Provides wireless connectivity throughout the office.
- **Firewall:** Protects the internal network from unauthorized access.
- **VPN Gateway:** Creates a secure connection between the branch office and headquarters.

4. Security Mechanisms

- Use a **site-to-site VPN** for encrypted communication with headquarters.
- Enable **firewall rules** to block unauthorized traffic.
- Use **WPA3/WPA2 encryption** for Wi-Fi.
- Apply strong passwords and user authentication.
- Keep router, firewall, and other network devices updated.
- Separate guest devices from company systems using different networks/VLANs.

5.Constraint-Based Decision

Constraint	Solution	Reason
No wired broadband	4G/5G	Available in many rural areas and easier to deploy
Limited bandwidth	Traffic prioritization/QoS	Gives video conferencing and cloud applications higher priority
Uninterrupted operation	Dual-WAN + backup link	Automatically switches when the main connection fails
Limited budget	4G/5G primary link	Lower installation cost than dedicated wired infrastructure
Security	Firewall + VPN	Protects communication and company data
Future growth	Scalable router, switch and Wi-Fi	Allows more users and devices to be added

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Constraint-Based Wireless Network Design for a University Campus

1. Problem Identification

The university needs to provide wireless internet access for **more than 3,000 students** across classrooms, laboratories, libraries, and hostels. The network must support many simultaneous users while maintaining good performance and security with a **limited number of Access Points (APs)**.

2. Constraints

- **User constraint:** More than 3,000 students may connect simultaneously.
- **Coverage constraint:** Classrooms, labs, libraries, and hostels must have reliable Wi-Fi coverage.
- **Interference constraint:** Overlapping signals must be minimized.
- **Performance constraint:** The network must handle high-density traffic.
- **Security constraint:** Only authorized students and staff should access the network.
- **Cost constraint:** The number of APs must be limited.
- **Scalability constraint:** The design should support future users and devices.

3. Proposed Wireless Network Design

Access Point Placement

Deploy APs based on **user density rather than simply geographical area**.

- Place more APs in high-density areas such as libraries, laboratories, and classrooms.
- Place APs at suitable ceiling positions for uniform coverage.
- Avoid placing APs near walls, metal objects, electrical equipment, or other sources of interference.
- Use a centralized wireless controller to monitor and manage APs.
- Use lower transmission power where necessary to reduce overlap and interference.

Frequency Band Selection

Use **5 GHz and 6 GHz bands as the primary bands** where supported by client devices.

- **5 GHz:** Provides higher capacity and less interference than 2.4 GHz.
- **6 GHz:** Provides additional clean spectrum and is useful for high-density areas with compatible devices.
- **2.4 GHz:** Keep mainly for older devices and devices requiring greater range.

Use automatic channel selection and non-overlapping channels to reduce interference.

4. Authentication and Security

Use **WPA3-Enterprise with 802.1X authentication** for students and staff.

A **RADIUS server** can authenticate users using their university credentials.

Separate networks can be created using VLANs:

- Student VLAN
- Faculty/Staff VLAN
- Guest VLAN
- Administrative VLAN

This prevents unauthorized users from directly accessing sensitive university systems.

5.Constraint-Based Analysis

Constraint	Solution	Benefit
3,000+ users	High-density AP deployment	Handles many simultaneous connections
Limited AP budget	APs placed according to user density	Avoids unnecessary APs
Signal interference	Channel planning and power control	Improves Wi-Fi performance
High bandwidth demand	5 GHz/6 GHz priority	Provides greater capacity
Security	WPA3-Enterprise + 802.1X	Strong user authentication
Different user groups	VLAN segmentation	Improves security and management
Future expansion	Centralized controller and scalable design	Easier to add APs later

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Constraint-Based Secure Network Architecture for a Financial Institution

1. Problem Identification

The financial institution needs a secure internal network that protects sensitive customer information and online banking services while maintaining good network performance and controlling operational costs.

2. Constraints

- **Security constraint:** Sensitive customer and financial data must be protected.
- **Performance constraint:** Security mechanisms should not significantly slow network traffic.
- **Compliance constraint:** The network must follow strict financial data security policies.
- **Cost constraint:** The solution should use existing infrastructure where possible.
- **Access constraint:** Only authorized employees should access sensitive information.
- **Availability constraint:** Banking services must remain available without unnecessary downtime.

3. Proposed Network Architecture

Network Segmentation

Use **VLANs and subnetting** to separate different departments and services.

- **Employee VLAN:** Normal employee systems.
- **Finance/Customer Data VLAN:** Systems handling sensitive customer information.
- **Server VLAN:** Internal application and database servers.
- **Guest VLAN:** Visitor devices with no access to internal systems.
- **DMZ:** Public-facing online banking and web services.

Access between VLANs should be controlled through Layer 3 switches and firewalls using strict access-control rules.

Firewall Deployment

Deploy a **Next-Generation Firewall (NGFW)** at the network perimeter and between critical internal network zones.

The firewall should provide:

- Access-control rules
- Application filtering
- Malware protection
- Traffic monitoring
- VPN security
- Network address translation

Only required ports and services should be allowed.

4. Authentication and Access Control

Use **Multi-Factor Authentication (MFA)** for employees accessing sensitive systems.

Combine:

- Username and strong password
- One-time password or authenticator
- Role-Based Access Control (RBAC)

Employees should receive only the permissions required for their jobs. This follows the **principle of least privilege**.

5. Intrusion Detection

Deploy **IDS/IPS** at important network points.

- **IDS** monitors network traffic and generates alerts for suspicious activity.
- **IPS** can automatically block detected malicious traffic.

Monitoring should focus especially on the server VLAN, customer-data VLAN, and DMZ.

6. Constraint-Based Analysis

Constraint	Solution	Benefit
High security requirement	VLANs, firewall, IDS/IPS	Reduces unauthorized access

Constraint	Solution	Benefit
Sensitive customer data	Isolated data/server VLANs	Limits attack exposure
Performance	Traffic filtering and optimized rules	Avoids unnecessary processing
Compliance	MFA, logging, access control	Supports security auditing
Limited budget	VLANs using existing switches	Reduces hardware costs
Unauthorized access	MFA + RBAC	Ensures only authorized users access data
Online banking security	DMZ + firewall	Separates public services from internal systems